Inch-Pound

A-A-55697
15 January 1995
SUPERSEDING
L-T-790B
April 14, 1976
AND
INT. AMENDMENT-1
March 11, 1977

COMMERCIAL ITEM DESCRIPTION

TUBE, NONMETALLIC; POLYVINYL CHLORIDE (PVC) FLEXIBLE (FOR LABORATORY AND MEDICAL USE)

The General Services Administration has authorized the use of this Commercial Item Description (CID).

1. **SCOPE.** This CID covers clear flexible PVC tubing intended for use in medical and laboratory procedures i.e., including but not limited to transfusions and infusions.

2. **CLASSIFICATION.**

2.1 Size and wall thickness. Plastic tubing shall be of the nominal size and wall thickness, as specified (see Table 1).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. or any data which may improve this document should be sent to: Commander, Defense Construction Supply Center, ATTN DCSC-VSS, P.O. Box 3990, Columbus, OH 43216-5000.

AMSC N/A

FSC 4720

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TABLE 1. Inside/outside diameter and thickness of tubes.

Size Code	* Nominal inside diameter		Outside diameter		** Nominal wall thickness	
	(inch)	(mm)	(inch)	(mm)	(inch)	(mm)
01	0.1250	(3.2)	0.1875	(4.8)	0.0313	(0.8)
02	0.1563	(4.0)	0.2813	(7.1)	0.0625	(1.6)
03	0.1875	(4.8)	0.3125	(7.9)	0.0625	(1.6)
04	0.2500	(6.4)	0.3750	(9.5)	0.0625	(1.6)
05	0.3125	(7.9)	0.4375	(10.5)	0.0625	(1.6)
06	0.3750	(9.5)	0.5000	(12.7)	0.0625	(1.6)
07	0.3750	(9.5)	0.6250	(15.9)	0.1250	(3.2)
08	0.5000	(12.7)	0.7500	(19.1)	0.1250	(3.2)
09	1.0000	(25.4)	2.0000	(50.8)	0.5000	(12.7)

Notes: * Tolerance on inside diameter for all sizes shall be ±.0156 inch (0.4 mm).
** Tolerance on wall thickness for all sizes shall be minus 0 or plus .0156 inch (0.4 mm).
*** Sizes other than those shown in table 1 may be ordered by specifying dimentional data in lieu of PIN (see 7.2).

3. SALIENT CHARACTERISTICS.

3.1 General. Tubing provided in accordance with this CID shall be colorless, clear, flexible PVC.

3.2 Dimensioning. The values stated in inch-pound units are to be regarded as the standard. The metric values stated in parentheses throughout this CID are for informational purposes only and have been rounded to only one decimal place for convenience.

3.3 Lengths. Tubing shall be furnished in the length specified (see Table 2 and 7.1c).

TABLE 2. Tubing lengths.

Length Code	Length		
A	50	Ft	(15,240 mm)
B	100	Ft	(30,480 mm)
C	250	Ft	(76,200 mm)
D	500	Ft	(152,400 mm)
E	1000	Ft	(304,800 mm)

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3.4 Form. Tubes shall be furnished in the form specified (see Table 3 and 7.1c).

TABLE 3. Tubing forms.

Form Code	Form
1	Coils
2	Reels
3	Straight

3.5 Flexibility. Tubing shall be capable of bending around a mandrel equal to five times the Outside Diameter (OD) of the tube without collapsing or restricting flow.

3.6 Physical properties. Physical properties of the tubes shall be in accordance with Table 4.

TABLE 4. Physical properties.

PROPERTY	REQUIREMENT	TEST METHOD (ASTM)
Brittleness temperature, °C	- 46	D746
Specific gravity	1.15 - 1.80	D792
Tensile strength, p.s.i.	1800 - 2100	D638
Elongation, percent	300 min.	D638

3.7 Chemical resistance. Tubing shall not discolor, flake, crack or pit when tested in accordance with USP-NF Physicochemical Tests-Plastics, ASTM D1003, Standard Method of Test for Haze and Luminous Transmittance of Transparent Plastics, and ASTM E308, Tentative Method of Test for Luminous Reflectance, Transmittance, and Color of Materials..

3.8 Sterilization. Tubing shall be capable of withstanding sterilization by steam, ethylene oxide, radiation, or other method that meets the requirements of the USP-NF, sterility tests for transfusion and infusion assemblies and similar medical devices.

3.9 Heat resistance. Tubing shall be capable of withstanding temperatures of 93 ± 3 degrees Centigrade for one hour without visual signs of deterioration such as discoloring, flaking, cracking, pitting, hardening, or melting.

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3.10 Biological reactivity. Tubing shall fulfill the requirements of USP Class II plastics and the Association for the Advancement of Medical Instrumentation, AAMI/ANSI 10993-1.

3.11 Metal content. The use of Hazardous materials shall be eliminated or minimized where possible. Under no circumstance shall, when tested (see 5.2), the ash residue of the tubes contain more than five parts per million (ppm) of heavy metals such as lead.

3.12 Workmanship. Tubes shall be free from blemishes and flaws which mar their appearance or affect their serviceability.

4 REGULATORY REQUIREMENTS.

4.1 Regulations and Laws. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Contractor Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets this CID, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract or order.

5.2 Metal concentration. To determine conformance to 3.11, two sample tubes shall be selected at random from each lot (lot is defined as a continuous run of tubing from the same batch of raw material, and offered for delivery under one contract). From each sample accurately weigh a five gram sample into a one ounce Vycor crucible. Reduce to ash in a muffle furnace at 650 degrees Centigrade for eight hours. Let cool to room temperature and add 15 ml of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for two to three minutes, remove, let cool and filter through Whatman #2 filter paper into a 50 ml volumetric flask. Rinse the residue from the filter with deionized water and dilute to 50 ml. Prepare standard lead solutions of one, two, and five ppm lead. Determine the absorbencies of these solutions at the 283 millimicron wave length using a Hollow Cathode Pb lamp and 10x scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303 Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

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Calculation: $\frac{A \times F \times \text{dilution}}{W} = \text{ppm lead}$

A = Absorbance
F = Factor = $\frac{\text{(Concentration standard in ppm Pb)}}{\text{absorbance}}$
W = Sample weight in grams

The calculated factors of the three standards are averaged to give ppm/unit A.

6. **PACKAGING.**

6.1 Preservation, packaging, and packing. Preservation, Packaging, and Packing shall be in accordance with ASTM D3951 or as specified in the contract or order (see 7.1d).

6.2 Marking and labeling. Marking and labeling shall be commercial or as specified in the contract or order (see 7.1e).

7. **NOTES.**

7.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this CID.
b. Size and wall thickness (see Table 1)
c. Length and form required (see 3.3 and 3.4).
d. Applicable level of preservation, packaging, and packing (see 6.1).
e. Applicable marking and labeling (see 6.2).
f. Quantity of tubing required (see 3.3).

7.2 Part Identification Number (PIN). The following CID based PIN procedure is for Government purposes and does not constitute a requirement for contractors/suppliers. The PIN shall be constructed as follows:

(A 55697-05-A-1)

- Form (see Table 3)
- Length (see Table 2)
- Size Code (see Table 1, Size Code 05)
- Basic CID Identifier (number)
- Document Identifier (CID)

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(Example: A55697-05-A-1) = Clear, flexable PVC tube, .313 inch nominal ID and .188 inch nominal OD, with a .031 nominal wall thickness, furnished in a 50 foot length coil.

Additional sizes may be ordered by stipulating Table 1, 2, and 3 Data in lieu of PIN (see 2.1) in the procurement document. A copy of the table 1 data shall be forward to the Preparing Activity with a request to have the additional size(s) added to the CID for future acquisition.

7.3 ASTM documents. ASTM documents cited may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.

7.4 United States Pharmacopeial Convention, Inc. United States Pharmacopeia (U.S.P.) may be obtained by submitting requests to Printing Company, 20th and Northampton Streets, Easton, PA. 18042 or by contacting U.S.P customers services, 12601 Twinbrook Parkway, Rockville, MA 20852 (AC) 301-881-0666.

7.5 (AAMI/ANSI documents. AAMI/ANSI documents may be obtained from The Association for the Advancement of Medical Instrumentation, Suite 400, 3330 Washington Blvd., Arlington, VA 22202.

7.6 Other related documents. Other related documents (not requirements of this specification) are: International Standards Organization (ISO) 1135-4, for transfusion sets, and ISO 8536-4 for infusions sets.

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PREPARING ACTIVITY:

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CIVIL AGENCY COORDINATING ACTIVITIES:

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(Project Number 4720-0050)

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